MODELLING OF ADDITIVE MANUFACTURING – COMPRESSION MOLDING PROCESS USING COMPUTATIONAL FLUID DYNAMICS

Berin Šeta [1], Marco Brander[1], Md. Tusher Mollah[1], Vipin Kumar[2], Deepak Kumar Pokkalla[2], Seokpum Kim[2], and Jon Spangenberg[1]
[1]Department of Mechanical Engineering
Technical University of Denmark
2800 Kgs. Lyngby, Denmark
[2]Manufacturing Science Division
Oak Ridge National Laboratory
2350 Cherahala Blvd, TN 37932, United States

ABSTRACT
A computational fluid dynamics model has been developed to predict the behavior of a printed strands during a novel material extrusion additive manufacturing and compression molding process. While the traditional additive manufacturing process enables control over the fiber orientation within the part, it would also result in high void content. On the other hand, compression molding produces parts with low porosity levels at rapid processing cycle time but lacks control over the microstructure. The novel additive manufacturing – compression molding (AM-CM) integrating both these processes offers control over both the microstructure and porosity to manufacturing high performance composite parts. The numerical model developed here enables to analyze the effect of processing parameters on the behaviour of printed layer that is subsequently compressed and to determine the optimal printing parameters for high-performance composite part design.

INTRODUCTION
A novel large-scale “additive manufacturing – compression molding” (AM-CM) process to achieve high-performance composites was developed recently [1, 2]. The additively manufactured preforms were compression molded to reduce void content while maintaining fiber alignment and thereby significantly enhancing mechanical properties. Modeling the fluid flow during layer deposition and subsequent squeezing is necessary to design parts with improved interlayer contact, specified dimensions, and prescribed mechanical properties. The focus of this paper is to develop a computational model to investigate the influence of rheological properties and processing parameters on layer deposition and deformation during AM-CM process and to propose simplistic approach for the prediction of fiber orientation. Orientation of the fibers and fluid flow are in general two-way coupled problem. Fibers tend to orientate in the direction of the flow creating anisotropic rheological properties of the fluid when highly oriented in a certain direction. The anisotropic behaviour is a trademark of the deposition flows, where fibers are highly oriented along the nozzle and afterwards in the printing direction when placed on the substrate [3]. In order to accurately predict the fiber orientation, one has to use the micro/macroscopic model which would include fibers as the orientation tensor or lagrangian particles. In this work, the simplistic approach is proposed, by placing tracers in the fluid before compression. After compression, the paths of those tracers are examined and compared with the newly modified orientation tensor.

COMPUTATIONAL MODEL
The computational model for novel AM-CM process consists of two main parts. The first part is related to the simulation of the layer deposition onto the solid-substrate, while the second one represents squeezing of the previously deposited material. The first part of the model solves the incompressible and isothermal two-phase fluid flow, where one phase is represented as the deposited material, while the other phase is the surrounding air. Among different methods, the modelling has been done using the well established volume-of-fluid (VoF) method [4-8], where both phases are solved with a single momentum equation and their interface is tracked by a prescribed scalar α. The value of α represents whether the domain contains air or deposited material, where the values between 0 and 1 represent the interface between phases.

The continuity and momentum equations can be written as:

$$\nabla . \boldsymbol{u} = 0 \tag{1}$$

$$\rho\left(\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u}.\nabla \boldsymbol{u}\right) = \rho \boldsymbol{g} - \nabla p + \nabla . \boldsymbol{S}_T + f_{\sigma i} \tag{2}$$

where $\boldsymbol{u}$ represents the velocity, p is the the pressure, $\boldsymbol{g} = (0,0,-g)$ is the gravitational vector, t is time, $f_{\sigma i}$ is surface tension and $\boldsymbol{S}_T$ is the deviatoric stress tensor. In order to solve the transport of phases, the transport equation of α has to be solved as:

$$\frac{\partial \alpha}{\partial t} + \nabla(\alpha \boldsymbol{u}) = 0 \tag{3}$$

The rheology of the material in exemplary case shown in Figs. 1, 2 and 3 is modelled using a constant viscosity of 5000 Pa.s and a density of 1000 kg/m^3 while air is modelled as a fluid with viscosity of 1 Pa.s and density of 0.01 kg/m^3. The choice of air properties, which obviously is not standard, is made up in order to prevent excessive flow within the air, as this can significantly increase the simulation time. On the other hand, it must be ensured that the chosen values do not affect the flow of the deposited material.

NUMERICAL PROCEDURE

The novelty of numerical procedure proposed in this paper when compared with other deposition flows developed in OpenFOAM is mainly within presribed motion of the nozzle and the usage of overset methodology. Most of the models [9, 10] use prescribed motion of the substrate, which siginificantly raise the computational cost of the simulation, especially in the deposition of multiple layers. On the other hand, sole process of the compressive molding is modelled using the same unconventional approach which employs the finite volume method and overset meshes, where the movement is prescribed to the compression step. The link between those two processes is established by mapping results (mapFields), once the deposition flow is completed, onto initial mesh/time of the compressive molding process. The material is deposited with an extrusion speed of 33.4 mm/s, while the nozzle movement speed is 50 mm/s. The nozzle diameter is 25 mm, while the layer height is half of the nozzle's diamater, 12.5 mm. In Figure 1, it is possible to see the deposited material, using constant viscosity value. The blue colour represents the contour of the $\alpha = 0.5$, which is considered as the interface between the two existing phases. Once the material is deposited, it is compressed by the upper plate (solid), as shown in Figure 2.

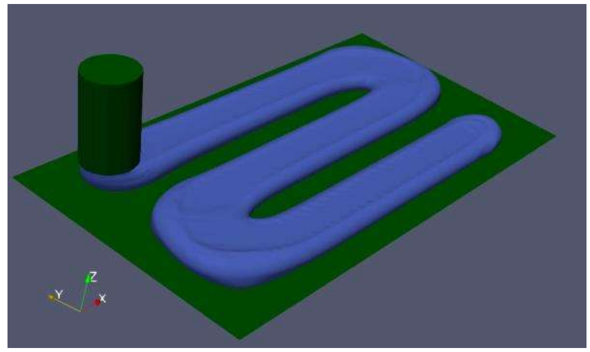

FIGURE 1. The deposited material (blue) on the substrate (green). First stage of the analysis.

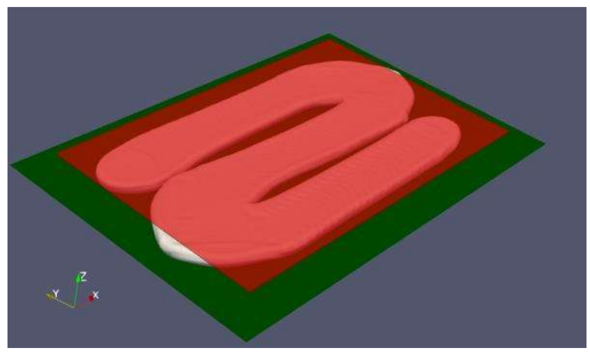

FIGURE 2. The deposited material (white) is compressed with plate (transparent red) on the substrate (green). Second stage of the analysis.

The upper plate (which represents a compression cycle) is moving downwards with the speed of 30 mm/s. The compression process transition is shown in Figure 3. The cross sections of the deposited material are shown through three different slices across the domain, and their deformation during the compression process points out in which way materials flows (Figure 3). Different rheological properties and printing processing parameters will influence the initial shape of the material, which later will play a significant role on the material location after compression [3].

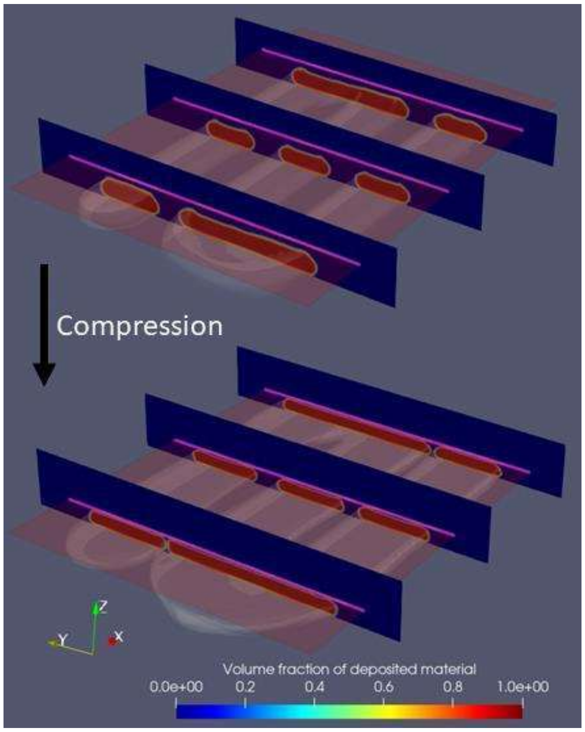

FIGURE 3. Compression process with slices alongside deposited material.

RESULTS

Along the study, multiple parameters were varied, including viscosity values when the Newtonian model was considered. The values varied from 1 Pa.s to 5000 Pa.s. The change was introduced only once the compression molding step started, as the deposition flow is calculated with the same processing parameters and rheological properties mentioned above. For the different values of constant viscosity simulated, no significant difference was observed in the way how strand is deformed and flow developed inside material. On the other hand, once the rheological model is changed to Bingham plastic, there was noticeable change in both shape of strand and established flow. In this work, Bingham plastic model is numerically defined using bi-viscous regularization, where the apparent high viscosity value is assigned to unyielded parts of the strand. The comparison between those two models is shown in Figure 4.

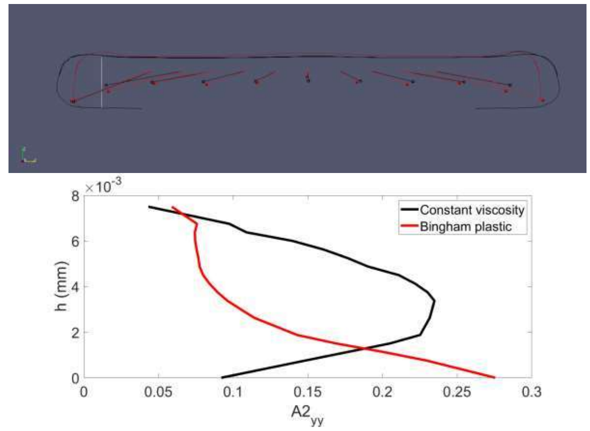

FIGURE 4. Comparison between model with constant viscosity (black) and Bingham plastic model (red). Initial position of tracers are at the origin of pathlines, while current position is represented as spheres. The outline of the strand represents isoline of phase function $\alpha = 0.5$ for both models. The bottom part represent the orientation tensor in the yy direction along the white line.

The tracers, represented as black/red spheres and their paths in time represented as black/red lines are visible in Figure 4. Knowing how particles were displaced in time, it is possible to estimate how fluid deformed in time, which means that this representation is superior compared to, for example, only velocity or strain rate tensor field that is fixed to a point in time. However, the trajectories of particles will not resemble orientation of the fibers, but rather gradient of particles displacement in certain direction could provide more insightful information about the fiber orientation. For example, in Figure 4, it is visible by looking carefully at pathlines, that around strand height center, black pathlines have higher gradient in y-direction (they diverge faster in the horizontal direction as compared to red ones). This suggests that the orientation of fibers in the yy direction around the strand height center should be higher in case of constant viscosity (black lines). The hypothesis is proven by plotting the yy component of the orientation tensor along the thin white vertical line. It is clear from Figure 4 (bottom part), that fibers are more oriented in the case with constant viscosity as predicted by the pathlines. In the Figure 5, it can be seen that pathlines of the spheres on the two top horizontal imaginary lines had higher gradient in the y direction, as compared to the spheres on the bottom imaginary horizontal line. This indicates that the probability of the tensor orientation changed more during compression in the top region, where the gradient of the horizontal component (y-axis) of the path trajectory was higher. This tendency is confirmed by looking at the actual orientation tensor field in the y direction along the vertical white line from Figure 5 (bottom part). It is noticeable that, for the initial random orientation field (dotted lines), indeed higher enhancement of the orientation in y direction is achieved in the top parts of the strand. The top part of Figure 5, represents the moment when the strand is compressed for 3 mm (red lines in the bottom part of Figure 5). Moreoever, the white isoline is the strand border while the colormap represents the yy component of the orientation tensor, with the initially solved orientation tensor during deposition flow (red solid line on bottom part of Figure 5).

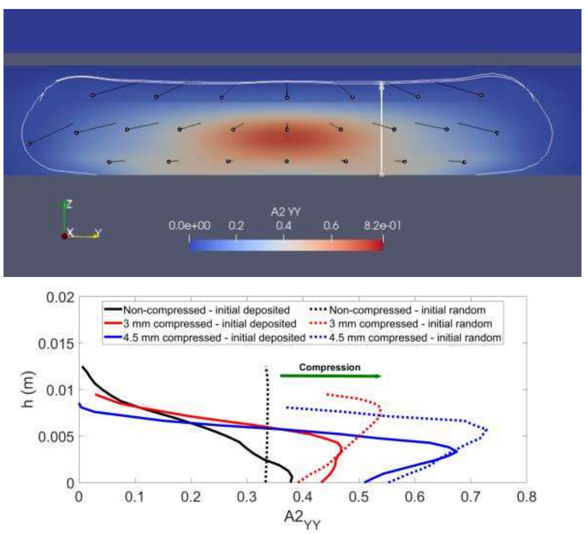

FIGURE 5. Component A2 YY (horizontal direction) of the fibers orientation tensor is presented as a colormap field in the top part. Black spheres /lines represent the pathlines of tracers during compression. Bottom part plots the A2 YY component along the thin white line for different levels of compression and different initial condition of fibers orientation.

The advantages of the pathline approach are that the results can be obtained very fast, and even retroactively (tracers can be introduced in the model once the simulation is finished if enough time data are stored). Pathline approach indirectly takes into account both vortricity and strain rate tensor, ensuring that their important role in macroscopic modelling of the fiber orientation is maintained [11]. However, it provides only qualitative estimation of the fiber orientation, without the possiblility to provide reliable quantification of the orientation probability in certain direction. By no means the approach takes into account the fiber-fiber interaction, aspect ratio and volume fraction of the fibers which are all important parameters when estimating the orientation tensor. For that reason, the simplistic approach is generally more appropriate in dilute suspension with short fibers, where the anistropic effect on rheology of the fluid is minimal along with fiber-fiber interactions. As shown above, using numerical modelling, it would be possible to determine favourable printing scenarios which will provide better interlayer contact and lower void content [12]. Moreover, the required amount of deposited material can be estimated in order to prevent deposition excessive quantity of material, which can affect the material properties after compression. Consequently, an optimized amount of the deposited material quantity will prevent the waste of material. Having in mind that thermoplastic is ocassionaly reinforced by fibers, and that the orientation of fibers highly influences the properties of the molded composite, numerical prediction of proper print strategies and the flow of the fluid while being compressed becomes a very relevant future task for the AM-CM process [13]. The proposed simplistic approach can be alternative to relatively complex and computationally expensive, potentially numerically unstable macroscopic models, when certain conditions, such as being in range of dilute suspensions, are met.

ACKNOWLEDGMENT

The authors would like to acknowledge the support of the Innovation Fund Denmark (Grant no. 0223-00084B).

REFERENCES

[1] Kumar V, Alwekar SP, Kunc V, Cakmak E, Kishore V, Smith T, Lindahl J, Vaidya U, Blue C, Theodore M, Kim S. High-performance molded composites using additively manufactured preforms with controlled fiber and pore morphology. Additive Manufacturing. 2021;37:101733.

[2] Kumar V, Nielson C, Yeole P, Spencer R, Cramer C, Badesha AS, Nuttall D, Vaidya U, Kunc V. Large-scale continuous carbon/glass fiber additive-compression molded composites. Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States); 2021

[3] Bertevas E, Férec J, Khoo BC, Ausias G, Phan-Thien N. Smoothed particle hydrodynamics (SPH) modeling of fiber orientation in a 3D printing process. Physics of Fluids. 2018;30(10):103103.
[4] Mollah MT, Comminal R, Serdeczny MP, Pedersen DB, Spangenberg J. Stability and deformations of deposited layers in material extrusion additive manufacturing. Additive Manufacturing. 2021; 46: 102193.
[5] Comminal R, Serdeczny MP, Pedersen DB, Spangenberg J. Numerical modeling of the strand deposition flow in extrusion-based additive manufacturing. Additive Manufacturing. 2018;20:68-76.
[6] Comminal R, Hattel JH, Spangenberg J. Numerical simulations of planar extrusion and fused filament fabrication of non-Newtonian fluids. Annu. Trans. Nord. Rheol. Soc.;25:263-70.
[7] Sandberg M, Yuksel O, Baran I, Hattel JH, Spangenberg J. Numerical and experimental analysis of resin-flow, heat-transfer, and cure in a resin-injection pultrusion process. Composites Part A: Applied Science and Manufacturing. 2021;143:106231.
[8] Serdeczny MP, Comminal R, Pedersen DB, Spangenberg J. Experimental validation of a numerical model for the strand shape in material extrusion additive manufacturing. Additive Manufacturing. 2018;24:145-53.
[9] Liu J, Anderson KL, Sridhar N. Direct simulation of polymer fused deposition modeling (fdm)—an implementation of the multi-phase viscoelastic solver in OpenFOAM. International Journal of Computational Methods. 2020;17(01):1844002.
[10] Wang Z, Smith DE. A Fully coupled simulation of planar deposition flow and fiber orientation in polymer composites additive manufacturing. Materials. 2021;14(10):2596.
[11] Advani SG, Tucker III CL. The use of tensors to describe and predict fiber orientation in short fiber composites. Journal of rheology. 1987;31(8):751-84.
[12] Pokkalla DK, Kumar V, Jo E, Hassen AA, Cakmak E, Alwekar S, Kunc V, Vaidya U, Baid HK, Kim S. Anisotropic mechanical properties of polymer composites from a hybrid additive manufacturing-compression molding process using x-ray computer tomography. InNondestructive Characterization and Monitoring of Advanced Materials, Aerospace, Civil Infrastructure, and Transportation XVI 2022 (Vol. 12047, pp. 319-328). SPIE.
[13] Serdeczny MP, Comminal R, Pederson DB, Spangenberg J. Numerical prediction of the porosity of parts fabricated with fused deposition modeling. In 2018 International Solid Freeform Fabrication Symposium. University of Texas at Austin.